Notes: STAT27400 Nonparametric Inference (W26)

Lecturer: Nikolaos Ignatiadis
Notes by: Aden Chen

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1 Overview

Our starting distribution is some distribution $\mathbb{P}$ from which we observe iid samples $Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} \mathbb{P}$, from which we hope to learn properties of $\mathbb{P}$. There are broadly two settings:

- "Unsupervised": $\mathbb{R}^p \ni Z_i = X_i \overset{\text{iid}}{\sim} \mathbb{P}$ are covariates.
- "Supervised": $\mathbb{R}^p \times \mathbb{R} \ni Z_i = (X_i, Y_i)$, where we think of X_i as covariates and Y_i as responses.

In either case, the workflow a nonparametric problem is the following:

(i) Determine the property of $\mathbb{P}$ we care about, $\theta := \theta(\mathbb{P})$.

(ii) How well did we do on this task? [Statistical decision theory]

(iii) What's the best we can do under given assumptions? [Information theory]

(iv) Propose a concrete method based on data at hand.

Example 1.1 (Step (i)). Suppose $\mathbb{R} \ni Z_i = Y_i \overset{\text{iid}}{\sim} \mathbb{P}$. We might care about:

- The mean $\theta := \mathbb{E}_{\mathbb{P}}[Y_i]$ (note that in writing this down we have already assumed that $\mathbb{P}$ has a finite first moment).
- The variance $\theta := \mathbb{V}_{\mathbb{P}}[Y_i] = \mathbb{E}_{\mathbb{P}}[(Y_i - \mathbb{E}_{\mathbb{P}}[Y_i]^2)]$.
- The median $\theta := \inf\{y \in \mathbb{R} : \mathbb{P}(Y_i \leq y) \geq 1/2\}$.
- The distribution in tails: $\theta(\mathbb{P}) := \mathbb{P}[Y_i \geq y | Y_i \geq t]$ for fixed $t, y \in \mathbb{R}$.
- The cumulative distribution function (CDF) $\theta(\mathbb{P}) := F(y) := \mathbb{P}(Y_i \leq y)$ for some fixed $y \in \mathbb{R}$.

Suppose $\mathbb{P}$ has density f. We may want to learn $\theta(\mathbb{P}) := f$ or $\theta(\mathbb{P}) := f(y_0)$ for some fixed $y_0 \in \mathbb{R}$. ☙

For now, let's suppose we want to learn $\theta(\mathbb{P}) := \mathbb{E}[\mathbb{P}]$ and walk through the other steps in the pipeline.

1.1 Example: Estimating the Mean

Example 1.2. A natural estimator is the sample mean (step (iv)):

$$\hat{\theta} := \frac{1}{n} \sum_{i=1}^{n} Y_i.$$

It is unbiased and consistent.

We may want to quantify how well we did (step (ii)) with the mean squared error (MSE): Assuming $\mathbb{E}[Y_i^2] < \infty$, we have with $\sigma^2 := \mathbb{V}(Y_i)$ that

$$\mathrm{MSE}(\hat{\theta}, \mathbb{P}) = \mathbb{V}(\hat{\theta}) = \frac{\sigma^2}{n}.$$

A lot of nonparametric problems are concerned with what the rate of convergence of the MSE is as $n \to \infty$. [Usually it is slower then n^{-1}.]

Finally, we may want to know what the best possible rate is (step (iii)).

Note that we used the following generic decomposition:

Proposition 1.3 (Bias-Variance Decomposition)**.** *For any estimator $\hat{\theta}$ of a parameter $\theta(\mathbb{P})$ such that $\mathbb{E}[\hat{\theta}^2] < \infty$, we have*

$$\mathrm{MSE}(\hat{\theta}, \mathbb{P}) := \mathbb{E}_{\mathbb{P}}[(\hat{\theta} - \theta(\mathbb{P}))^2] = \mathbb{V}(\hat{\theta}) + \big(\mathbb{E}[\hat{\theta}] - \theta)\big)^2.$$

Proof. Note that

$$\begin{aligned}
\mathbb{E}[(\hat{\theta} - \theta(\mathbb{P}))^2] &= \mathbb{E}\Big[\big(\hat{\theta} - \mathbb{E}[\hat{\theta}] + \mathbb{E}[\hat{\theta}] - \theta(\mathbb{P})\big)^2\Big] \\
&= \mathbb{E}\Big[\big(\hat{\theta} - \mathbb{E}[\hat{\theta}]\big)^2\Big] + 2\mathbb{E}\Big[\big(\hat{\theta} - \mathbb{E}[\hat{\theta}]\big)\big(\mathbb{E}[\hat{\theta}] - \theta(\mathbb{P})\big)\Big] + \mathbb{E}\Big[\big(\mathbb{E}[\hat{\theta}] - \theta(\mathbb{P})\big)^2\Big] \\
&= \mathbb{V}(\hat{\theta}) + 0 + \big(\mathbb{E}[\hat{\theta}] - \theta(\mathbb{P})\big)^2.
\end{aligned}$$

□

Remark 1.4 (Minimax). Consider a set of possible distributions $\mathcal{P}$. We may want to minimize the worst-case risk

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}[(\hat{\theta} - \theta(\mathbb{P}))^2].$$

Example 1.5. In the example above, we are implicating restricting our analysis to $\mathcal{P} := \{\mathbb{P} : \mathbb{V}_{\mathbb{P}}[Y_i] < \infty\}$. In this class of distributions, the worse-case MSE of the sample mean is unbounded. More assumptions on $\mathbb{P}$ are needed to get nontrivial worse-case risk.

For instance, restricting to $\mathcal{P}_M := \{\mathbb{P} : \mathbb{V}_{\mathbb{P}}[Y_i] \leq M\}$ for some fixed $M < \infty$, we have that

$$\sup_{\mathbb{P} \in \mathcal{P}_M} \mathrm{MSE}(\hat{\theta}, \mathbb{P}) = \frac{M}{n}.$$

In fact,

$$\inf_{\hat{\theta}} \sup_{\mathbb{P} \in \mathcal{P}_M} \mathrm{MSE}(\hat{\theta}, \mathbb{P}) = \frac{M}{n}.$$

That is, the sample mean is minimax optimal (step (iii)). We may think of this as a two player game, where the statistician chooses $\hat{\theta}$ and nature chooses $\mathbb{P} \in \mathcal{P}_M$.

1.2 Example: Estimating the CDF

Example 1.6. Fix $y \in \mathbb{R}$. Suppose we are interested in $\theta := \mathbb{P}[Y_i \le y] = F(y)$. A natural estimator is the **empirical CDF (ECDF)** at y:

$$\hat{F}_n(y) := \hat{\theta} := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{Y_i \le y\}.$$

Note that this is precisely the sample mean if we write

$$\theta := \mathbb{P}[Y_i \le y] = \mathbb{E}[\mathbb{1}\{Y_i \le y\}].$$

By the previous analysis, we have that

$$\mathrm{MSE}(\hat{\theta}, \mathbb{P}) = \frac{F(y)(1 - F(y))}{n}$$

Let $\mathcal{P}$ be the class of all distributions on $\mathbb{R}$. We have

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathrm{MSE}(\hat{\theta}, \mathbb{P}) = \frac{1}{4n}.$$

Note that the MSE is small when $F(y)$ is close to 0 or 1, and largest when $F(y) = 1/2$. This is more an artifact of the choice of loss function (MSE), which depends on the scale of θ.

For the two objects we have considered so far, $\theta(\mathbb{P}) := \mathbb{E}[Y_i]$ and $\theta(\mathbb{P}) := F(y)$, the estimators that are good in the worst-case sense have 0 bias and $\mathrm{MSE} = \mathbb{V} = O(n^{-1})$. In parametric problems this is the typical case (even when there is bias, we typically have $n \cdot \mathrm{Bias}^2 \to 0$). However, in nonparametric problems the optimal estimators often have nonzero bias and $\mathbb{V} \approx \mathrm{Bias}^2$, with slower convergence rates.

An informal description: the harder the problem, the more important the bias.

A common approach to obtaining a good estimator is to consider with a family of estimators $\hat{\theta}_\lambda \in \Lambda$ and pick λ so that the bias-variance tradeoff is optimized.

1.3 Example: Estimating the Density

Example 1.7 (Density Estimation). Suppose $\mathbb{P}$ has density f and we want to estimate $\theta := f(y_0)$. For simplicity, assume $Y_i \subset [0, 1]$. We may partition $[0, 1]$ into k intervals $I_1, \ldots, I_K$ of length $h := K^{-1}$. We may consider $\hat{\theta} := \mathrm{Histogram}(y_0; K) := (nh)^{-1} \sum \mathbb{1}\{y_i \in I_j\}$, where $y_0 \in I_j$.

Here, the tuning parameter is K (or equivalently h). Intuitively, for larger h, the variance decreases but the bias increases.

We formalized this intuition: Note that

$$\mathbb{V}(\hat{\theta}_k) = \frac{\mathbb{P}[Y_i \in I_j](1 - \mathbb{P}[Y_i \in I_j])}{nh^2} \le \frac{\mathbb{P}[Y_i \in I_j]}{nh^2}.$$

Let's suppose (Assumption 1) that $f(y) \le M < \infty$ for each y. Then,

$$\mathbb{V}(\hat{\theta}_k) \le \frac{Mh}{nh^2} = \frac{M}{nh}.$$

As we send $h \to 0$, the variance blows up.

Next, we analyze the bias. Additional assumptions much be imposed. Indeed, without additional structure, f may not even be defined. A typical strategy here is to "bet on smoothness." Let's assume (Assumption 2) that f is **L-Lipschitz**:

$$|f(y) - f(y')| \le L|y - y'|, \quad \forall y, y'.$$

Now, writing $[a_j, b_j] := I_j$, we have

$$\hat{\theta}_k(y_0) = \frac{\sum \mathbb{1}\{y_i \in I_j\}}{nh} = \frac{\mathbb{P}[Y_i \in I_j]}{h} = \frac{F(b_j) - F(a_j)}{h},$$

and so the bias is

$$|\mathrm{Bias}(\hat{\theta}_k)| = \left|\frac{F(b_j) - F(a_j)}{h} - f(y_0)\right| = \left|f(y_j) - f(y_0)\right| \le L|y_j - y_0| \le Lh.$$

From the first equality we see that as we send $h \to 0$, the bias goes to 0. The second equality follows from the mean value theorem.

Combining the estimates for bias and variance, we have

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathrm{MSE}_{\mathbb{P}}(\hat{\theta}_k, \mathbb{P}) \le \frac{M}{nh} + L^2 h^2,$$

which can be optimized by choosing h. It turns out it suffices to balance variance and squared bias: $M/nh = L^2 h^2$, which gives (ignoring constants) $1/nh = h^2$ and then $h = n^{-1/3}$. With this choice of h, we have

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathrm{MSE}_{\mathbb{P}}(\hat{\theta}_k, \mathbb{P}) \le C(L, M) \cdot n^{-2/3}.$$

We write

$$\mathrm{MSE}(\hat{\theta}_k, \mathbb{P}) \lesssim_{M,L} n^{-2/3}.$$

This is a common nonparametric rate.

It turns out that a minimax result holds, suggesting that we cannot do much better:

$$\inf_{\hat{\theta}} \sup_{\mathbb{P} \in \mathcal{P}} \mathrm{MSE}_{\mathbb{P}}(\hat{\theta}, \mathbb{P}) \ge c(L, M) \cdot n^{-2/3}.$$

Remark 1.8. What if we care about $f(\cdot)$. A suitable risk function is the integrated mean squared error (IMSE):

$$\mathrm{IMSE}(\hat{f}, \mathbb{P}) := \mathbb{E}\left[\int_0^1 (\hat{f}(y) - f(y))^2 w(y)\, \mathrm{d}y\right],$$

where we often pick the uniform weight function $w(y) \equiv 1$.

Does out previous analysis extend to this setting? No, since out previous bounds do not depend on y_0.

$$\mathrm{IMSE}(\hat{f}, \mathbb{P}) := \int_0^1 \mathbb{E}[(\hat{f}(y) - f(y))^2]\, \mathrm{d}y \le Cn^{-2/3}.$$

Q: how to form confidence intervals for $f(y_0)$?

The classical approach $\hat{\theta} \pm 1.96 \operatorname{se} \hat{\theta}$ can fail in nonparametric setting. The standard argument that justifies this construction of the CI is that $(\hat{\theta} - \theta)/\operatorname{se} \hat{\theta} \xrightarrow{\mathrm{d}} \mathcal{N}(0, 1)$. In the nonparametric setting, however, we typically only have $(\hat{\theta} - \mathbb{E}\hat{\theta})/\operatorname{se} \hat{\theta} \xrightarrow{\mathrm{d}} \mathcal{N}(0, 1)$, but $\mathbb{E}\hat{\theta} \neq \theta$ due to bias.

One of the main approaches to this problem is **undersmoothing**: pick tuning parameters so that bias is negligible compared to standard error. Another main approach is **bias correction**: estimate the bias and subtract it off. (**Robust bias correction**: Calonico, Cattaneo, and Farrell Econometrica.) The third main approach is **bias-aware inference**: we inflate the CI to account for the bias. The fourth approach is to ignore the bias and report CIs for $\mathbb{E}\hat{\theta}$ instead of θ.

2 Nonparametric Regression

The setting is $Z_i = (X_i, Y_i) \overset{\text{iid}}{\sim} \mathbb{P}$, where $X_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}$. We want to predict Y_i from X_i.

We hope to find function $m : \mathbb{R}^d \to \mathbb{R}$ that minimizes

$$\mathbb{E}[(Y_i - m(X_i))^2].$$

It turns out that the optimal function is the **regression function (conditional expectation function)** $m^*(x) = \mathbb{E}[Y_i | X_i = x]$.

Proof.

$$\begin{aligned}\mathbb{E}[(Y_i - m(X_i))^2] &= \mathbb{E}[(Y_i - m^*(X_i) + m^*(X_i) - m(X_i))^2] \\ &= \mathbb{E}[(Y_i - m^*(X_i))^2] + 2\mathbb{E}[(Y_i - m^*(X_i))(m^*(X_i) - m(X_i))] \\ &\quad + \mathbb{E}[(m^*(X_i) - m(X_i))^2],\end{aligned}$$

The first term does not depend on m, the last term is minimized when $m = m^*$, and the cross term is 0 from the law of iterated expectation:

$$\mathbb{E}[(Y_i - m^*(X_i))(m^*(X_i) - m(X_i))] = \mathbb{E}\Big[\mathbb{E}[Y_i - m^*(X_i)|X_i](m^*(X_i) - m(X_i))\Big] = 0.$$

□

How do we estimate m^*? Suppose we observe $\mathcal{D}_{\text{train}} = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$. A few approaches:

(i) **Regressogram**: Partition the covariate space into bins and use the average response in each bin as the prediction for points in that bin.

(ii) Impose explicit modelling assumptions: $m(x) = m(x; \theta)$, or e.g., $m(x) = \sum b_j \psi_j(x)$, where ψ_j are splines.

(iii) Any supervised learning method: e.g., deep learning (random forests, boosting).

Nonparametric regression is a problem we are really good at solving (wer are better than our understanding of why). One flavor of modern nonparametric is to take a black-box for nonparametric regression and ask how we can repurpose it for other statistical tasks.

If we make assumptions such as smoothness, how well can we do?

Example 2.1. Suppose $X \in \{1, \ldots, K\}$ and $\mathbb{V}(Y_i | X = k) \le \sigma^2$ for some $\sigma > 0$ and each k. One estimator is the sample mean within each bin:

$$\hat{m}(k) = \begin{cases} \frac{1}{\#\{j : X_j = k\}} \sum_{j : X_j = k} Y_j, & \#\{j : X_j = k\} > 0 \\ 0, & \text{otherwise} \end{cases}.$$

Morally, we have

$$\text{Bias}(\hat{m}(k)) = 0, \quad \mathbb{V}(\hat{m}(k)) \lesssim \frac{\sigma^2}{n_k}.$$

This is not quite correct since (1) when $n_k = 0$ the estimator is 0, and (2) $\#\{j : X_j = k\}$ is a random quantity. One way to resolve this is via conditioning:

$$\mathrm{Bias}(\hat{m}(x)|X_{1:n}) = \begin{cases} 0, & n_k > 0 \\ -m(k), & n_k = 0 \end{cases}, \quad \mathbb{V}(\hat{m}(k)|X_{1:n}) = \begin{cases} \frac{\sigma^2}{n_k}, & n_k > 0 \\ 0, & n_k = 0 \end{cases}.$$

This is sometimes called a **fixed-X regression**. Thus we have

$$\mathrm{MSE}(\hat{m}(k)) = \mathbb{E}\Big[\mathrm{Bias}^2(\hat{m}(k)|\mathcal{D}_{\text{train}})\Big] + \mathbb{E}\Big[\mathbb{V}(\hat{m}(k)|X_{1:n})\Big].$$

The regressogram is a similar case. But even if $\#\{X_i \in I_k\} > 0$, the bias is nonzero unless m is constant on I_k. One way to control the bias is to assume smoothness of m, say by assuming m is L-Lipschitz.

2.1 The k-NN (k-Nearest Neighbors) Estimator

We want to estimate $m(x_0)$. Define

$$N_k(x_0) := \{i \in \{1, \dots, n\} : |X_i - x_0| \text{ is among } k \text{ smallest}\}.$$

Note that this is not well-defined if there are ties. There are various ways to break ties, but let's assume that there are no ties by assuming $X_i \sim \mathrm{Uniform}[0,1]^d$.

Definition 2.2. The **kNN estimator** of $m(x_0)$ is

$$\hat{m}_{k\mathrm{NN}}(x_0) := \frac{1}{k} \sum_{i \in N_k(x_0)} Y_i.$$

Example 2.3. 1-NN: $\hat{m}(X_i) = Y_i$, thus $\sum (Y_i - \hat{m}(X_i))^2 = 0$. Thus 1-NN is sometimes called an **interpolating** predictor. The traditional view is that this is a bad predictor since it overfits. The modern view says not necessarily.

Note that we have **boundary bias** at the endpoints since the neighborhood is one-sided there. This can be a problem in e.g., regression discontinuity designs, where the point of interest is at the boundary.

As k increases, the bias increases and the variance decreases. When k is too large, we have **underfitting/oversmoothing**. When k is too small, we have **overfitting/undersmoothing**.

Let's make this formal.

Assumption 2.4.

(i) $\mathbb{V}(Y_i|X_i = k) \le \sigma^2$ for some $\sigma > 0$ and each k.

(ii) m is L-Lipschitz.

(iii) $X_i \sim \mathrm{Uniform}[0,1]^d$.

We start with the bias:

$$
\begin{aligned}
|\mathrm{Bias}(\hat{m}(x_0)|X_{1:n})| = |\mathbb{E}[\hat{m}(x_0)|X_{1:n}] - m(x_0)| &= \left| \sum_{i \in N_k(x_0)} \frac{m(Y_i)}{k} - m(x_0) \right| \\
&\leq \frac{1}{k} \sum_{i \in N_k(x_0)} |m(Y_i) - m(x_0)| \\
&\leq L \|X_{(k)}(x_0) - x_0\|,
\end{aligned}
$$

where $X_{(k)}(x_0)$ is the k-th nearest neighbor of x_0 among $X_1, \ldots, X_n$.

Now the conditional variance:

$$
\mathbb{V}(\hat{m}(x_0)|X_{1:n}) = \frac{1}{k^2} \sum_{i \in N_k(x_0)} \mathbb{V}(Y_i|X_i) \leq \frac{\sigma^2}{k}.
$$

We now can combine these two estimates:

$$
\mathrm{MSE}(\hat{m}(x_0)) \leq \frac{\sigma^2}{k} + L^2 \mathbb{E}\left[\|X_{(k)}(x_0) - x_0\|^2 \right].
$$

Now suppose $d = 1$ and thus $X_i \sim \mathrm{Uniform}[0, 1]$. It turns out that

$$
\mathbb{E}\left[\left\| X_{(k)}(x_0) - x_0 \right\|^2 \right] \approx \left(\frac{k}{n} \right)^2.
$$

More generally, in dimension d we have

$$
\mathbb{E}\left[\left\| X_{(k)}(x_0) - x_0 \right\|^2 \right] \approx \left(\frac{k}{n} \right)^{2/d},
$$

a "curse of dimensionality."

Proof. We start by proving this for $k = 1$. Fix $x_0 \in [0, 1]^d$. Then, $\mathbb{P}\left[\|X - x_0\| \leq \varepsilon \right] \geq$

$c_d\varepsilon^d$ (the ball $B_\varepsilon(x_0)$ has volume $c_d\varepsilon^d$). Now consider the nearest neighbor $X_{(1)}(x_0)$:

$$\begin{aligned}
\mathbb{E}\left[\|X_{(1)}(x_0)-x_0\|^2\right] &= \int_0^\infty \mathbb{P}\left[\|X_{(1)}(x_0)-x_0\|^2 \geq \varepsilon\right]\ \mathrm{d}\varepsilon \\
&= \int_0^\infty \mathbb{P}\left[\|X_j - x_0\| \geq \sqrt{\epsilon}, \forall j\right]\ \mathrm{d}\varepsilon \\
&= \int_0^\infty \prod_j \mathbb{P}\left[\|X_j - x_0\| \geq \sqrt{\varepsilon}\right]\ \mathrm{d}\varepsilon \\
&= \int_0^\infty \left(1-\mathbb{P}\left[\|X-x_0\| \leq \sqrt{\varepsilon}\right]\right)^n\ \mathrm{d}\varepsilon \\
&\leq \int_0^\infty \left(1-c_d\varepsilon^{d/2}\right)^n\ \mathrm{d}\varepsilon \\
&\leq \int_0^\infty \exp\left(-nc_d\varepsilon^{d/2}\right)\ \mathrm{d}\varepsilon \\
&= \int_0^\infty n^{-2/d}c_d^{-2/d}\exp(-u^{d/2})\ \mathrm{d}u \\
&\lesssim \left(\frac{1}{n}\right)^{2/d},
\end{aligned}$$

where the last line is from the change of variables $u = n^{2/d}c_d^{2/d}\varepsilon$.

check proof

We extend this from 1-NN to k-NN: suppose n is a multiple of $2k$ and partition $\{1,\ldots,n\}$ unto $2k$ equal blocks $I_1,\ldots,I_{2k}$. We use the notation $X_{(k)}(x_0;I)$ to denote the kth NN of x_0 in I. Since $X_{(1)}(x_0;I_j) \geq X_{(k)}(x_0,\{1,\ldots,n\})$ in at least k blocks, we have

$$\|X_{(k)}(x_0,\{1,\ldots,n\})-x_0\|^2 \leq \frac{1}{k}\sum_j^{2k}\|X_{(n)}(x_0,I_j)-x_0\|^2.$$

Thus,

$$\begin{aligned}
\mathbb{E}\left[\|X_{(k)}(x_0)-x_0\|^2\right] &\leq \frac{1}{k}\sum_j^{2k}\mathbb{E}\|X_{(n)}(x_0,I_j)-x_0\|^2 \\
&\leq \frac{2k}{k}\cdot\left(\frac{2k}{n}\right)^{2/d} \\
&\lesssim \left(\frac{k}{n}\right)^{2/d}.
\end{aligned}$$

□

Thus,

$$\mathrm{MSE}(\hat{m}(x_0)) \lesssim_{L,\sigma^2} \frac{1}{k}+\left(\frac{k}{n}\right)^{2/d}.$$

What is the optimal rate we can achieve by picking k?

Again, we need just equate the variance and squared bias:

$$\frac{1}{k} = \left(\frac{k}{n}\right)^{2/d} \implies k^{1+2/d} = n^{2/d} \implies k = n^{2/(2+d)}.$$

This gives the rate

$$\mathrm{MSE}(\hat{m}(x_0)) \lesssim_{L,\sigma^2} n^{-2/(2+d)}.$$

Recall that in the density estimation problem with L-Lipschitz densities we had rate $n^{-2/3}$, which matches this when $d = 1$.

Remark 2.5 (Some intuition for rates). For a rate of n^{-1}, we need to double the sample size to halve the error. For a rate of $n^{-1/2}$, we need to quadruple the sample size to halve the error. ☕

Did we need $X_i \sim \mathrm{Uniform}[0,1]^d$? Only in claiming $\mathbb{P}[\|X_i - x_0\| \le \varepsilon] \ge c_d \varepsilon^d$. This suggest the following relaxation: If x_i has density f on $[0,1]^d$ and suppose $f(x) \ge c > 0$ for all $x \in [0,1]^d$, then the same rate holds, with the caveat that the MSE will depend on c.

What if $\inf f(x) = 0$? Then our analysis do not apply, but there is still hope. Let's consider the integrated MSE:

$$\mathbb{E}[(\hat{m}(X) - m(X))^2] = \int \mathbb{E}[(\hat{m}(x) - m(x))^2] f(x)\, \mathrm{d}x.$$

Suppose $f(x_0) \approx 0$, then $\mathbb{E}[(\hat{m}(x_0) - m(x_0))]$ can be large without affecting the integrated MSE too much. Thus it turns out that we can still get the same rate for the integrated MSE without lower bounding the density:

Theorem 2.6. *Suppose $X_i \in [0,1]^d$, $\mathbb{V}[y_i | X_i = x] \le \sigma^2$ for all x, and m is L-Lipschitz. For optimally tuned k-NN, we have for $d \ge 3$ that*

$$\mathbb{E}[(\hat{m}(X) - m(X))^2] \lesssim_{L,\sigma^2} n^{-\frac{2}{2+d}}.$$

(Not true for $d = 1, 2$.)

In fact we have the much cooler

Theorem 2.7 (Universal Consistency)**.** *Let $(X_i, Y_i) \overset{\text{iid}}{\sim} \mathbb{P}$, $\mathbb{E}[Y_i^2] < \infty$, and suppose k is such that $k/n \to 0$, $k \to \infty$. Then*

$$\mathbb{E}[(\hat{m}_{kNN}(X) - m(X))^2] \longrightarrow 0.$$

Note that we do not need to impose smoothness assumptions, but also do not get convergence rates, or uniform convergence.

2.2 Alternative Assumptions

We need stronger assumptions to do well in higher dimensions (with these assumptions, k-NN may or may not be rate optimal).

2.2.1 Higher Order Smoothness

Higher order smoothness are often imposed using **Hölder classes**.

Definition 2.8. Denote $\lfloor \beta \rfloor$ as the largest integer strictly less than β.[1] The **Hölder class** $\Sigma(\beta, L)$, $\beta > 0$ contains all functions m such that m is $\lfloor \beta \rfloor$-times differentiable and for all $x, x' \in \mathbb{R}^d$,

$$\left| m^{(\lfloor \beta \rfloor)}(x) - m^{(\lfloor \beta \rfloor)}(x') \right| \leq L \|x - x'\|^{\beta - \lfloor \beta \rfloor}.$$

Example 2.9.

- $\Sigma(1, L)$ is the class of L-Lipschitz functions.
- $\Sigma(2, L)$ is the class of functions with L-Lipschitz first derivatives.

It turns out that the minimax rates are roughly

$$n^{-\frac{2\beta}{2\beta+d}} = n^{-\frac{2s}{2s+1}}, \quad s := b/d.$$

The larger β is, the faster the rate. As $s \to \infty$, the rate becomes closer to n^{-1}. In general, k-NN does not achieve this rate.

2.2.2 Structural Assumptions

Some examples:

- Additive models:
$$m(x) = \sum_{j=1}^{d} m_j(x_j),$$
where each m_j is, say, L-Lipschitz. Sometimes we can get rates as if in the univariate case.
- $m(\cdot)$ looks like a neural network.

2.2.3 Shape Assumptions

Examples:

- Monotonicity. In the one dimensional case, assuming $m(\cdot)$ is increasing gives a rate of $n^{-2/3}$ even without smoothness assumptions.
- Convexity.
- Log-concavity.

This is usually analyzed for losses such as $\mathbb{E}[(\hat{m}(X) - m(X))^2]$, where X follows some distribution (the integrated MSE).

[1] This is not the usual definition of floor. Note for example that $\lfloor 1 \rfloor = 0$.

2.2.4 The Manifold Hypothesis

Informally: In d dimension, the data may lie on or near a lower dimensional manifold, say with an "effective" dimension of s. Assuming Lipschitz continuity, Kpotufe (?) '11 shows that k-NN achieves the rate $n^{-\frac{2}{2+s}}$ (compare this to $n^{-\frac{2}{2+d}}$). In this case, k-NN adapts to geometry.

3 Kernel Methods

Local average:

$$\hat{m}(x_0) := \frac{\sum_{i:|X_i - x_0| \le h} y_i}{\#\{i : |X_i - x_0| \le h\}}$$

This is the special case of the Nadaraya-Watson estimator with uniform kernel $k(u) = \mathbb{1}\{|u| \le 1\}/2$.

Nadaraya-Watson: defines a kernel $k(\cdot)$ ($k(0)$ large, $k(u)$ small for large u) and given k, h, defines

$$\hat{m}(x) := \frac{\sum_{i=1}^n k\left(\frac{X_i - x}{h}\right) Y_i}{\sum_{i=1}^n k\left(\frac{X_i - x}{h}\right)}.$$

A local weighted average.

These kernels are density functions and symmetric around 0. Sometimes they have compact support $[-1, 1]$.

Q: when is $\hat{m}$ continuous in x? If $k(\cdot)$ is continuous and denominator is nonzero.

Before studying NW, let's study an analogous idea for density estimation, called **kernel density estimation** (KDE). Suppose $K(\cdot)$ is a kernel with $k \ge 0$ and $\int K(u)\,\mathrm{d}u = 1$. Let's assume also that K is bounded and supported on $[-1, 1]$ (these two conditions are not needed but they simplify the analysis). Let

$$\hat{f}(x_0) := \frac{1}{nh} \sum_i K\left(\frac{X_i - x_0}{h}\right).$$

Think of this as adding a bump of shape K at each data point.

If $\hat{f}(x_0)$ consistent? Suppose f is continuous at x_0, and x_0 lies in the interior of $[0, 1]$ so there exists $H > 0$ such that $[x_0 - H, x_0 + H] \subset [0, 1]$.

Let's start with computing

$$\begin{aligned}\mathbb{E}\hat{f}(x_0) &= \frac{1}{h}\mathbb{E}\left[K\left(\frac{X_i - x_0}{h}\right)\right] = \frac{1}{h}\int K\left(\frac{z - z_0}{h}\right) h(z)\,\mathrm{d}z \\ &= \frac{1}{h} \cdot h \int K(u) f(x_0 + uh)\,\mathrm{d}u = \int_{-1}^{1} K(u) f(x_0 + uh)\,\mathrm{d}u.\end{aligned}$$

In the second line we use the change of variables $u = (z - x_0)/h$. As $h \to 0$, the integral converges to $f(x_0)$ by continuity of f and the fact that K is a density. The bias converges to 0 as $h \to 0$.

Next, we compute the variance: A similar argument as that used in the analysis of the histogram estimator shows that

$$\mathbb{V}(\hat{f}(x_0)) \lesssim \frac{1}{nh} \longrightarrow 0.$$

This gives consistency of $\hat{f}(x_0)$ when $h \to 0$ and $nh \to \infty$.

Note the boundary bias: when x_0 is near 0 or 1, the above analysis fails since the support of K extends beyond $[0, 1]$. In fact KDE (the vanilla version of it) satisfies

$$\hat{f}(0) \xrightarrow{\text{p}} \frac{1}{2} f(0).$$

Boundary bias is much worse than for nonparametric regression.